

The Two-Base Integer-Exponent Problem

An Independent Proof Audit, Unimodular Gap Arithmetic,
Rational Secant Approximants, and a Fixed-Order Integer-Jet Barrier

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Abstract

The Alaoglu–Erdős conjecture asks whether a real number x satisfying $2^x, 3^x \in \mathbb{Z}$ must be an integer. The attached source report was audited line by line at the level of its mathematical claims. It contains a large body of exact reductions, formalized partial results, conditional structure, and failed or incomplete research programs, but it explicitly does not contain an unconditional proof. The present document is deliberately nonduplicative: it does not reproduce that corpus. Instead it gives an independent proof ledger and develops four additions.

First, a general unimodular collision theorem controls common divisors of binomial gaps $B^q - A^p$. Applied to consecutive continued-fraction convergents to $\log 3 / \log 2$, it proves that the gaps $3^{q_j} - 2^{p_j}$ are pairwise coprime at adjacent indices. Second, under a hypothetical counterexample, the corresponding output gaps produce positive rational divided differences and rational secant approximants to x that alternate rigorously around x ; their reduced denominator supports obey the same unimodular separation. Third, a finite order-signature theorem shows that no finite collection of strict monomial comparisons can distinguish the pair $(2, 3)$ from infinitely many unrelated coprime integer pairs. Fourth, a matrix divisibility theorem proves that fixed-order Euler or Hasse integer jets contribute only $O(RJ \log L)$ to the logarithmic arithmetic budget of an $R \times R$ determinant with exponent scale L and jet order J ; therefore fixed-order jets cannot alter an R^2 leading constant when $\log L = o(R)$.

These results are rigorous and new relative to the attached report, but they do not close the final algebraic-independence gap. No unconditional proof of the Alaoglu–Erdős conjecture is claimed. The exact missing implication is isolated, and a complete conditional proof from the Four Exponentials Conjecture is included.

Outcome and claim ledger**Unconditional proof of the full conjecture.**

Not obtained; no such proof is present in the attached report.

Conditional proof from Four Exponentials.

Complete and elementary once the conjectural input is granted.

Counterexample restrictions.

Any counterexample is positive, irrational, transcendental, and gives multiplicatively independent integer outputs.

New arithmetic theorem.

Unimodular exponent pairs force a sharp gcd formula for binomial gaps.

New approximation theorem.

Hypothetical output gaps give rational secant approximants alternating around x , with separated denominator support.

New no-go theorem.

Finitely many monomial order comparisons cannot characterize the special logarithmic slope.

New determinant barrier.

Fixed-order entrywise integer jets are lower-order at the R^2 scale unless $J \log L \gtrsim R$ or a genuinely global cancellation mechanism is found.

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1 The problem and the standard boundary

1.1 Statement

Conjecture 1.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Because 2^x and 3^x are positive, the integer hypothesis means that they are positive integers. Set

$$M := 2^x, \quad N := 3^x, \quad (1)$$

whenever a hypothetical solution is under discussion, and put

$$\alpha := \frac{\log 3}{\log 2} = \log_2 3. \quad (2)$$

Then $N = M^\alpha$, and the conjecture is equivalently the assertion that the only positive integers M for which M^α is an integer are powers of 2.

1.2 Why a source audit matters

The attached source file

`/Article/alaoglu-erdos-unified-report/alaoglu-erdos-unified-report.tex`

is a very large unified research report. Its title page and several later status statements explicitly say that the full conjecture remains open. It contains many exact lemmas, conditional classifications, determinant constructions, computational exclusions, and proposed interfaces, but the existence of such a large body of correct partial work is not itself a proof. The decisive audit question is whether any section derives a contradiction from the sole assumptions in theorem 1.1. It does not.

The present report therefore follows two rules.

- (i) It does not copy or paraphrase at length the attached report’s existing twenty-part survey, exact conditional monoid classification, determinant hierarchies, p -adic transport calculations, special-function audits, or formalization catalogue.
- (ii) Every new claim below is proved from explicitly stated hypotheses, and every point at which a known transcendence conjecture would be needed is marked as such.

2 Elementary reductions that every proof must respect

Proposition 2.1 (Sign and small interval). *If $2^x \in \mathbb{Z}$, then either $x = 0$ or $x \geq 1$. In particular, a nonintegral solution of theorem 1.1 would satisfy $x > 1$.*

Proof. If $x < 0$, then $0 < 2^x < 1$, so 2^x cannot be a positive integer. If $0 < x < 1$, then $1 < 2^x < 2$, again impossible. The remaining possibilities are $x = 0$ and $x \geq 1$. A nonintegral solution cannot equal 1. $\square$

Proposition 2.2 (Rational exponents). *Let $a > 1$ be an integer. If $x \in \mathbb{Q}$ and $a^x \in \mathbb{Z}$, then the denominator of x in lowest terms divides every prime valuation of a . In particular, if $a = 2$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Proof. Write $x = r/s$ with $r \in \mathbb{Z}$, $s \in \mathbb{N}$, and $\gcd(r, s) = 1$. The case $r < 0$ is excluded when a^x is a positive integer other than zero. Put $a^x = m \in \mathbb{N}$. Then $m^s = a^r$. For every prime p ,

$$s v_p(m) = r v_p(a).$$

Since $\gcd(r, s) = 1$, one has $s \mid v_p(a)$ for every $p \mid a$. For $a = 2$, $v_2(a) = 1$, hence $s = 1$. $\square$

Corollary 2.3. *Every nonintegral solution of theorem 1.1 is irrational.*

Proposition 2.4 (Algebraic exponents). *Every nonintegral solution of theorem 1.1 is transcendental.*

Proof. By theorem 2.3, such an x is irrational. If it were algebraic, then the Gelfond–Schneider theorem would make 2^x transcendental, contrary to $2^x \in \mathbb{Z}$. $\square$

Proposition 2.5 (Output multiplicative independence). *Let $x > 0$ and let M, N be as in (1). Then M and N are multiplicatively independent: if $M^r N^s = 1$ for $r, s \in \mathbb{Z}$, then $r = s = 0$.*

Proof. The relation gives $2^{xr} 3^{xs} = 1$, hence $2^r 3^s = 1$ after taking the positive x th root. Unique factorization gives $r = s = 0$. $\square$

2.1 Equivalent logarithmic formulations

For positive integers M, N , the equality

$$\frac{\log M}{\log 2} = \frac{\log N}{\log 3}$$

is equivalent to

$$\det \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log N \end{pmatrix} = 0. \quad (3)$$

Thus the problem asks for a very special classification of rank-one 2×2 matrices of logarithms of positive integers. Baker’s theorem controls nonzero *linear* forms in logarithms with algebraic coefficients. Equation (3) is instead a bilinear relation among four logarithms. Treating the coefficients $\log M$ and $\log N$ as if they were algebraic is precisely the invalid step in several tempting arguments.

3 The exact Four Exponentials implication

Conjecture 3.1 (Four Exponentials). *Let u_1, u_2 be linearly independent over $\mathbb{Q}$, and let v_1, v_2 be linearly independent over $\mathbb{Q}$. Then at least one of the four numbers $\exp(u_i v_j)$, $1 \leq i, j \leq 2$, is transcendental.*

Theorem 3.2 (Conditional resolution). *The Four Exponentials Conjecture implies the Alaoglu–Erdős conjecture.*

Proof. Assume 2^x and 3^x are integers. If $x \notin \mathbb{Q}$, take

$$(u_1, u_2) = (\log 2, \log 3), \quad (v_1, v_2) = (1, x).$$

The first pair is $\mathbb{Q}$ -linearly independent because 2 and 3 are multiplicatively independent; the second pair is $\mathbb{Q}$ -linearly independent because x is irrational. Yet all four exponentials

$$e^{\log 2} = 2, \quad e^{\log 3} = 3, \quad e^{x \log 2} = 2^x, \quad e^{x \log 3} = 3^x$$

are algebraic, contradicting theorem 3.1. Therefore $x \in \mathbb{Q}$, and theorem 2.2 gives $x \in \mathbb{Z}$. $\square$

The unavoidable status conclusion

The argument above is complete, but its decisive input is the unproved Four Exponentials Conjecture. Replacing that conjecture by a sentence such as “the four exponentials are algebraic, hence the rows or columns are dependent” is not a proof; it is exactly the conjectural implication. Any unconditional solution must either prove this special case of Four Exponentials or exploit the integrality of the four exponentials in a way not available for arbitrary algebraic values.

4 Unimodular arithmetic of binomial gaps

The first new contribution is a general gcd theorem for gaps $B^q - A^p$. It is elementary, but it organizes a substantial amount of local arithmetic in one determinant parameter.

4.1 The collision determinant

Definition 4.1. For exponent pairs $(p, q), (r, s) \in \mathbb{N}^2$, define their determinant distance by

$$\Delta((p, q), (r, s)) := ps - rq, \quad D := |ps - rq|.$$

The pairs are *unimodular neighbors* when $D = 1$.

Theorem 4.2 (Unimodular gap gcd theorem). *Let $A, B \geq 2$ be coprime integers, let $p, q, r, s \in \mathbb{N}$, and assume $D = |ps - rq| > 0$. Then*

$$\gcd(B^q - A^p, B^s - A^r) \mid \gcd(A^D - 1, B^D - 1). \quad (4)$$

In particular, if $D = 1$, then the sharp identity

$$\gcd(B^q - A^p, B^s - A^r) = \gcd(A - 1, B - 1) \quad (5)$$

holds.

Proof. Let d be the gcd on the left of (4). Since $\gcd(A, B) = 1$, d is coprime to AB : a prime dividing both d and A would also divide B^q , and similarly with A and B interchanged. Hence A and B are units modulo d . We have

$$B^q \equiv A^p \pmod{d}, \quad B^s \equiv A^r \pmod{d}.$$

Raise the first congruence to the power s and the second to the power q . The left sides agree, so

$$A^{ps} \equiv A^{rq} \pmod{d},$$

and therefore $A^{ps-rq} \equiv 1 \pmod{d}$. The same argument, raising the first congruence to r and the second to p , gives $B^{rq-sp} \equiv 1 \pmod{d}$. Inverting when necessary yields $A^D \equiv B^D \equiv 1 \pmod{d}$, proving (4). If $D = 1$, this shows that the left-hand gcd divides $\gcd(A - 1, B - 1)$. Conversely, every divisor of both $A - 1$ and $B - 1$ divides each gap $B^q - A^p$ because $A \equiv B \equiv 1$ modulo that divisor. This proves the equality (5). $\square$

Remark 4.3 (Primewise version without coprimality). If A and B are not assumed coprime, the same proof applies to every prime $\ell \nmid AB$. Thus the part of the left-hand gcd supported away from AB divides $\gcd(A^D - 1, B^D - 1)$. Equivalently, any such common prime satisfies

$$\text{ord}_\ell(A) \mid D, \quad \text{ord}_\ell(B) \mid D.$$

4.2 The continued-fraction recurrence shadow

The determinant proof gives the cleanest gcd statement. There is also an exact recurrence explaining why adjacent gcds propagate unchanged along a continued fraction.

Proposition 4.4 (Gap recurrence). *Suppose exponent pairs satisfy*

$$p_{j+1} = a_{j+1}p_j + p_{j-1}, \quad q_{j+1} = a_{j+1}q_j + q_{j-1}$$

with $a_{j+1} \in \mathbb{N}$. Put $h_j = B^{q_j} - A^{p_j}$. Then

$$h_{j+1} = B^{q_{j-1}}h_j \sum_{t=0}^{a_{j+1}-1} B^{q_j(a_{j+1}-1-t)} A^{p_j t} + A^{a_{j+1}p_j} h_{j-1}. \quad (6)$$

If $\gcd(A, B) = 1$, then

$$\gcd(h_{j+1}, h_j) = \gcd(h_j, h_{j-1}). \quad (7)$$

Proof. Substitute the recurrences for p_{j+1}, q_{j+1} and add and subtract $B^{q_{j-1}} A^{a_{j+1}p_j}$:

$$\begin{aligned} h_{j+1} &= B^{q_{j-1}}(B^{q_j})^{a_{j+1}} - A^{p_{j-1}}(A^{p_j})^{a_{j+1}} \\ &= B^{q_{j-1}}((B^{q_j})^{a_{j+1}} - (A^{p_j})^{a_{j+1}}) + A^{a_{j+1}p_j}(B^{q_{j-1}} - A^{p_{j-1}}). \end{aligned}$$

Factor the difference of powers to obtain (6). Modulo h_j this gives

$$h_{j+1} \equiv A^{a_{j+1}p_j} h_{j-1} \pmod{h_j}.$$

When $\gcd(A, B) = 1$, every $h_j = B^{q_j} - A^{p_j}$ is coprime to A . Multiplication by $A^{a_{j+1}p_j}$ therefore does not change the gcd with h_j , proving (7). $\square$

Corollary 4.5 (Exact adjacent gcd along the 2–3 slope). *Let (p_j/q_j) be the convergents to $\log 3/\log 2$. For every coprime $A, B \geq 2$ and every $j \geq 0$,*

$$\gcd(B^{q_j} - A^{p_j}, B^{q_{j+1}} - A^{p_{j+1}}) = \gcd(A - 1, B - 1).$$

Proof. Consecutive convergents are unimodular, so this is already (5). Alternatively, theorem 4.4 shows the adjacent gcd is constant; at the first two convergents $(1, 1)$ and $(2, 1)$ it is

$$\gcd(B - A, B - A^2) = \gcd(A - 1, B - 1),$$

where coprimality removes the possible factor from A . $\square$

4.3 Continued fractions and adjacent coprimality

Let

$$\frac{p_j}{q_j} = [a_0; a_1, \dots, a_j]$$

be the convergents to $\alpha = \log 3/\log 2$, indexed from $j = 0$. Consecutive convergents satisfy

$$|p_j q_{j+1} - p_{j+1} q_j| = 1. \quad (8)$$

Define the signed base gaps

$$g_j := 3^{q_j} - 2^{p_j}. \quad (9)$$

Theorem 4.6 (Adjacent base gaps are coprime). *For every $j \geq 0$,*

$$\gcd(g_j, g_{j+1}) = 1. \quad (10)$$

Proof. Apply theorem 4.2 with $(A, B) = (2, 3)$ and the two consecutive convergent pairs. By (8), $D = 1$, so the gcd divides $\gcd(2 - 1, 3 - 1) = 1$. $\square$

Corollary 4.7 (Higher-distance collision spectrum). *For any $j \neq k$, put*

$$D_{jk} := |p_j q_k - p_k q_j|.$$

Then

$$\gcd(g_j, g_k) \mid \gcd(2^{D_{jk}} - 1, 3^{D_{jk}} - 1).$$

Thus every repeated prime divisor of two base gaps has multiplicative orders of both 2 and 3 dividing the determinant distance D_{jk} .

The first convergents and gaps are shown in table 1. The huge growth of the gaps is compatible with their adjacent coprimality; relative Diophantine closeness does not mean small absolute difference once the exponential scale is restored.

Table 1: Initial convergents to $\log 3 / \log 2$ and signed gaps.

j	p_j	q_j	$\text{sgn}(g_j)$	$ g_j $
0	1	1	+	1
1	2	1	−	1
2	3	2	+	1
3	8	5	−	13
4	19	12	+	7153
5	65	41	−	420491770248316829
6	84	53	+	40432553845953101497907
7	485	306	−	a 144-digit integer
8	1054	665	+	a 313-digit integer

4.4 Output gaps under a hypothetical counterexample

Assume now that $x > 1$ is a nonintegral solution, and retain $M = 2^x$, $N = 3^x$. Define

$$G_j := N^{q_j} - M^{p_j}. \quad (11)$$

The common logarithmic slope implies

$$\text{sgn}(G_j) = \text{sgn}(g_j). \quad (12)$$

Indeed, if

$$\delta_j := q_j \log 3 - p_j \log 2, \quad (13)$$

then

$$g_j = 2^{p_j} (e^{\delta_j} - 1), \quad G_j = M^{p_j} (e^{x\delta_j} - 1).$$

Corollary 4.8 (Output collision restriction). *For consecutive convergents, if $\gcd(M, N) = 1$, then the exact formula*

$$\gcd(G_j, G_{j+1}) = \gcd(M - 1, N - 1)$$

holds. Without the coprimality assumption, the part of $\gcd(G_j, G_{j+1})$ supported away from MN satisfies the same divisibility. More generally, at indices $j \neq k$ every common prime $\ell \nmid MN$ has $\text{ord}_\ell(M), \text{ord}_\ell(N) \mid D_{jk}$.

Proof. Apply theorem 4.2 or theorem 4.3 to $(A, B) = (M, N)$. $\square$

Remark 4.9 (What this does not prove). The base gaps have adjacent gcd exactly 1. When M, N are coprime, the output gaps have the equally rigid but not contradictory constant adjacent gcd $\gcd(M-1, N-1)$; without coprimality, additional factors supported on MN may occur. There is no known theorem forcing the same prime to recur in adjacent output gaps. Consequently the gcd theorem supplies arithmetic separation, not a contradiction.

5 Transported divided differences

The same convergents produce a rational family whose analytic behavior is completely rigid.

5.1 Positive rational divided differences

For any positive p, q with $3^q \neq 2^p$, define

$$Q_{p,q} := \frac{N^q - M^p}{3^q - 2^p}. \quad (14)$$

Theorem 5.1 (Exact divided-difference transport). *Under a hypothetical solution $x > 1$, $Q_{p,q}$ is a positive rational number and*

$$Q_{p,q} = x \int_0^1 ((1-t)2^p + t3^q)^{x-1} dt. \quad (15)$$

If $Q_{p,q} = a_{p,q}/b_{p,q}$ in lowest terms with $b_{p,q} > 0$, then

$$b_{p,q} = \frac{|3^q - 2^p|}{\gcd(|3^q - 2^p|, |N^q - M^p|)}, \quad b_{p,q} \mid |3^q - 2^p|. \quad (16)$$

Proof. Both numerator and denominator in (14) are nonzero integers. Their signs agree because $t \mapsto t^x$ is strictly increasing on the positive reals, so the quotient is positive. The identity

$$\frac{b^x - a^x}{b - a} = x \int_0^1 ((1-t)a + tb)^{x-1} dt$$

follows from the fundamental theorem of calculus after the affine substitution $u = (1-t)a + tb$; take $a = 2^p$, $b = 3^q$. Formula (16) is the standard reduction of a ratio of two integers. $\square$

For convergents, write $Q_j := Q_{p_j, q_j}$ and let b_j be its reduced denominator.

Corollary 5.2 (Adjacent denominator separation). *For every $j \geq 0$,*

$$\gcd(b_j, b_{j+1}) = 1.$$

Proof. By theorem 5.1, $b_j \mid |q_j|$. Apply theorem 4.6. $\square$

5.2 Asymptotics at convergent nodes

The continued-fraction estimate gives

$$|\delta_j| = \log 2 \, |q_j \alpha - p_j| < \frac{\log 2}{q_{j+1}}. \quad (17)$$

Moreover, the signs of δ_j alternate.

Proposition 5.3 (Asymptotic secant scale). *As $j \rightarrow \infty$,*

$$Q_j = 2^{p_j(x-1)} \frac{e^{x\delta_j} - 1}{e^{\delta_j} - 1} \quad (18)$$

$$= x 2^{p_j(x-1)} \left(1 + \frac{x-1}{2} \delta_j + O_x(\delta_j^2) \right) \quad (19)$$

$$\log Q_j = (x-1)p_j \log 2 + \log x + O_x(q_{j+1}^{-1}) \quad (20)$$

Proof. Use $3^{q_j} = 2^{p_j} e^{\delta_j}$ and $N^{q_j} = M^{p_j} e^{x\delta_j}$ in (14). Since $M^{p_j}/2^{p_j} = 2^{p_j(x-1)}$, this gives (18). Taylor expansion at $\delta = 0$ yields

$$\frac{e^{x\delta} - 1}{e^{\delta} - 1} = x + \frac{x(x-1)}{2} \delta + \frac{x(x-1)(2x-1)}{12} \delta^2 + O_x(\delta^3).$$

Combine this with (17); taking logarithms is legitimate for all sufficiently large j because the parenthetical factor tends to 1. $\square$

5.3 Rational approximants that bracket the unknown exponent

Normalize away the exponential scale by defining

$$T_j := \frac{2^{p_j}}{M^{p_j}} Q_j = \frac{2^{p_j}(N^{q_j} - M^{p_j})}{M^{p_j}(3^{q_j} - 2^{p_j})}. \quad (21)$$

Every T_j is rational. Define the analytic secant function

$$\Phi_x(\delta) := \begin{cases} \frac{e^{x\delta} - 1}{e^{\delta} - 1}, & \delta \neq 0, \\ x, & \delta = 0. \end{cases} \quad (22)$$

Then $T_j = \Phi_x(\delta_j)$.

Theorem 5.4 (Alternating rational secant brackets). *Let $x > 1$. The function Φ_x is strictly increasing on $\mathbb{R}$. Hence*

$$\operatorname{sgn}(T_j - x) = \operatorname{sgn}(\delta_j), \quad (23)$$

and consecutive T_j lie on opposite sides of x . Quantitatively,

$$T_j - x = \frac{x(x-1)}{2} \delta_j + O_x(\delta_j^2) = O_x(q_{j+1}^{-1}). \quad (24)$$

Thus a hypothetical counterexample generates an explicit alternating sequence of rational upper and lower bounds converging to x .

Proof. For every real δ ,

$$\Phi_x(\delta) = x \int_0^1 (1 - t + te^{\delta})^{x-1} dt. \quad (25)$$

Differentiating under the integral sign gives

$$\Phi'_x(\delta) = x(x-1)e^{\delta} \int_0^1 t(1 - t + te^{\delta})^{x-2} dt > 0.$$

Therefore $\Phi_x(\delta) - \Phi_x(0)$ has the sign of δ , proving (23). The expansion in (24) is the Taylor series already used in theorem 5.3, and the final estimate follows from (17). $\square$

Definition 5.5 (Denominator away from a fixed integer). For a positive integer d and $M \geq 2$, let $d^{(M)}$ denote the largest divisor of d coprime to M .

Corollary 5.6 (External denominator separation for the rational brackets). *Let d_j be the reduced denominator of T_j . Then*

$$d_j^{(M)} \mid |g_j|, \quad \gcd(d_j^{(M)}, d_{j+1}^{(M)}) = 1.$$

Proof. Equation (21) is obtained from Q_j by multiplying by $2^{p_j}/M^{p_j}$. New denominator primes can therefore come only from M ; away from M , the reduced denominator still divides the denominator b_j of Q_j , hence divides g_j . Now use theorem 4.6. $\square$

5.4 Why the rational brackets are not yet an irrationality-measure contradiction

The approximation error in (24) is of order q_{j+1}^{-1} , whereas the denominator of T_j can contain M^{p_j} and has exponential height in p_j . In terms of that denominator, the approximation is far too weak to contradict any general irrationality measure. More importantly, x is already forced to be transcendental by theorem 2.4, so algebraic irrationality measures do not apply. The new information is not the existence of rational approximants, but their canonical construction, alternation, and prime-support separation.

6 A finite order-signature no-go theorem

The common logarithmic slope means that every monomial comparison is transported:

$$2^a 3^b < 2^c 3^d \iff M^a N^b < M^c N^d.$$

It is tempting to search for a finite list of such inequalities that only the pair (2, 3) can satisfy. The next theorem rules out that strategy in its pure form.

Theorem 6.1 (Finite order-signature indistinguishability). *Let $F \subset \mathbb{Z}^2 \setminus \{(0, 0)\}$ be finite. There exist infinitely many coprime, multiplicatively independent integer pairs (M, N) with $M, N > 1$ such that*

$$\operatorname{sgn}(a \log M + b \log N) = \operatorname{sgn}(a \log 2 + b \log 3) \quad \text{for every } (a, b) \in F. \quad (26)$$

Proof. Write $\rho = \log N / \log M$. Since $\alpha = \log 3 / \log 2$ is irrational, $a + b\alpha \neq 0$ for every nonzero $(a, b) \in \mathbb{Z}^2$. Because F is finite, there is an open interval I containing α on which every function $\rho \mapsto a + b\rho$ has the same sign as at α .

Choose an arbitrarily large prime P and set $M = P$. Among any two consecutive integers nearest to P^α , at least one is not divisible by P ; choose such an integer as N . Then $\gcd(M, N) = 1$, and

$$\frac{N}{P^\alpha} \longrightarrow 1, \quad \frac{\log N}{\log P} \longrightarrow \alpha \quad (P \rightarrow \infty).$$

For all sufficiently large choices, $\rho \in I$, so

$$\operatorname{sgn}(a \log M + b \log N) = \operatorname{sgn}(\log M) \operatorname{sgn}(a + b\rho) = \operatorname{sgn}(a + b\alpha),$$

which is the right side of (26). The pair is multiplicatively independent because two coprime integers greater than 1 cannot satisfy a nontrivial integer power relation. Infinitely many primes P give infinitely many pairs. $\square$

Corollary 6.2 (Finite monomial order patterns). *For every finite set $E \subset \mathbb{Z}^2$, there exist infinitely many coprime, multiplicatively independent pairs (M, N) for which the entire strict order pattern of the monomials $M^a N^b$, $(a, b) \in E$, is identical to that of $2^a 3^b$.*

Proof. Apply theorem 6.1 to all nonzero differences of pairs in E . $\square$

Warning 6.3. This theorem does not rule out a finite proof using congruences, exact integrality, valuations, or transcendence input. It rules out only a proof whose distinguishing information is a finite collection of strict monomial order comparisons. Any order-theoretic approach must use an infinite or uniform property, or combine order with genuinely arithmetic data.

7 A fixed-order integer-jet barrier

Integer jets are attractive in determinant methods because Euler derivatives preserve integrality at integer points:

$$(X\partial_X)^r (Y\partial_Y)^s (X^n Y^k) = n^r k^s X^n Y^k.$$

The attached report identifies nonvanishing and scaling as unresolved. The next result gives a sharp general budget for the automatic divisibility that can come from the entrywise integer multipliers alone.

7.1 A matrix theorem

For an integer matrix $C = (c_{ij})_{1 \leq i, j \leq R}$, allow zero entries and set $v_p(0) = +\infty$. Let

$$\Sigma(C) := \{\sigma \in S_R : c_{i, \sigma(i)} \neq 0 \text{ for all } i\}.$$

If $\Sigma(C) \neq \emptyset$, define

$$\tau_p(C) := \min_{\sigma \in \Sigma(C)} \sum_{i=1}^R v_p(c_{i, \sigma(i)}), \quad T(C) := \prod_p p^{\tau_p(C)}. \quad (27)$$

Only finitely many factors in $T(C)$ are nontrivial.

Theorem 7.1 (Entrywise decoration budget). *Let $E = (e_{ij})$ and $C = (c_{ij})$ be integer $R \times R$ matrices. Assume $\Sigma(C) \neq \emptyset$ and $|c_{ij}| \leq B$ for every nonzero entry, where $B \geq 1$. Then*

$$T(C) \mid \det(E \circ C), \quad (28)$$

where $\circ$ denotes entrywise product, and

$$\log T(C) \leq R \log B. \quad (29)$$

If additionally $|e_{ij}| \leq E_0$, then

$$\log |\det(E \circ C)| \leq \log(R!) + R \log E_0 + R \log B \quad (30)$$

whenever the determinant is nonzero.

Proof. Expand the determinant by permutations. A term with a zero c -entry vanishes. Every nonzero term indexed by $\sigma \in \Sigma(C)$ has p -adic valuation at least

$$\sum_i v_p(c_{i, \sigma(i)}) \geq \tau_p(C),$$

because the e_{ij} are integers. Hence every term, and therefore their sum, is divisible by $p^{\tau_p(C)}$ for every p , proving (28).

For the size bound, use “sum of minima is at most the minimum of sums”:

$$\begin{aligned} \log T(C) &= \sum_p \tau_p(C) \log p \\ &\leq \min_{\sigma \in \Sigma(C)} \sum_p \sum_i v_p(c_{i,\sigma(i)}) \log p \\ &= \min_{\sigma \in \Sigma(C)} \sum_i \log |c_{i,\sigma(i)}| \leq R \log B. \end{aligned}$$

Finally, every Leibniz term of $\det(E \circ C)$ has absolute value at most $(E_0 B)^R$, and there are $R!$ terms, giving (30). $\square$

Remark 7.2 (Optimality of the scale). The bound $R \log B$ cannot be improved in general: take $C = BI_R$ and an appropriate diagonal E . The theorem is therefore a structural ceiling, not merely a loose use of Hadamard’s inequality.

7.2 Euler and Hasse jets

Suppose exponents n, k lie in $[0, L] \cap \mathbb{Z}$ and all Euler jet orders obey $r + s \leq J$. The multiplier satisfies

$$|n^r k^s| \leq L^{r+s} \leq L^J$$

when nonzero.

Corollary 7.3 (Euler-jet budget). *For an $R \times R$ determinant obtained by entrywise decorating an integral evaluation matrix with Euler-jet multipliers of total order at most J and exponent scale at most L , the automatic extra divisor supplied solely by those multipliers has logarithm at most*

$$RJ \log L. \tag{31}$$

The same quantity is the maximum added term in the crude logarithmic archimedean bound.

Proof. Apply theorem 7.1 with $B = L^J$. $\square$

Hasse derivatives produce multipliers $\binom{n}{r} \binom{k}{s}$, also bounded by $L^{r+s} \leq L^J$. They additionally lower the powers of X and Y ; those row or column factors must be charged separately and cannot be counted as a free jet gain.

Corollary 7.4 (Leading-order no-go for fixed-order jets). *Consider a sequence of $R \times R$ determinant constructions with exponent scale L_R and jet order J_R . If*

$$J_R \log L_R = o(R), \tag{32}$$

then the termwise guaranteed integer-jet divisor and the corresponding crude archimedean multiplier budget are both $o(R^2)$ in logarithmic scale. In particular, fixed J_R with polynomially growing L_R cannot alter an R^2 leading constant in an archimedean-versus-nonarchimedean determinant comparison.

Proof. By (31), the contribution is at most $RJ_R \log L_R = o(R^2)$. $\square$

What a successful jet method must add

At least one of the following must occur:

- (a) $J_R \log L_R \gtrsim R$, so that the jet order or exponent scale is genuinely large;
- (b) the determinant has systematic cancellation beyond the minimum valuation of its individual Leibniz terms;
- (c) confluence creates a global factorization or rank defect not representable as bounded entrywise integer decoration;
- (d) a nonlocal arithmetic identity couples several primes or several jet levels at once.

Fixed-order differentiation alone cannot supply the missing leading-order arithmetic constant.

8 Stress tests of possible closure arguments

This section records the last steps one might hope to extract from the new lemmas and explains precisely why each step is currently unavailable.

8.1 Can adjacent coprimality force a contradiction?

No. The integers g_j are allowed to introduce new prime divisors at every index. The theorem $\gcd(g_j, g_{j+1}) = 1$ is entirely compatible with their exponential growth. The reduced denominators b_j of Q_j inherit this separation, but a sequence of rationals can have pairwise coprime adjacent denominators indefinitely. A contradiction would require a separate theorem forcing denominator recurrence or confinement to a fixed finite prime set. No such theorem follows from the hypotheses.

8.2 Can one transfer primitive primes from base gaps to output gaps?

The equality of logarithmic slopes synchronizes the signs and relative sizes of g_j and G_j , but it does not synchronize their residues modulo a prime. A prime dividing g_j satisfies $3^{q_j} \equiv 2^{p_j}$, whereas a prime dividing G_j satisfies $N^{q_j} \equiv M^{p_j}$. The real-logarithmic relation gives no direct congruence between these two statements. A theorem forcing infinitely many common prime divisors between corresponding base and output gaps would be a genuinely new local-global principle and would go well beyond the elementary gcd calculations.

8.3 Can the rational brackets prove x rational?

The approximants T_j are canonical and alternate, but their error is only $O(q_{j+1}^{-1})$ while their possible denominators are exponentially large. They do not give a Liouville-type approximation. Nor can Roth's theorem or an algebraic irrationality measure be applied, because a counterexample is already known to be transcendental.

8.4 Can Padé approximation close the logarithm-product relation?

A Padé construction becomes decisive only if its denominator clearing and archimedean smallness beat each other at the same leading scale. The relation (3) involves products of logarithms, not a nonzero linear form with algebraic coefficients. Recent Padé advances for polylogarithmic periods provide useful techniques, but they do not currently imply the required special algebraic independence of $\log 2, \log 3, \log M, \log N$.

8.5 Can automaticity or Cobham-type rigidity be invoked?

Cobham theorems require an actual finite automaton, recognizability in two multiplicatively independent numeration systems, or an equivalent finite-state hypothesis. The integrality of 2^x and 3^x does not by itself make the digit sequence of any associated quantity automatic. Inferring automaticity from the existence of two integer values is an unsupported bridge.

8.6 Can finite order geometry identify the slope?

No finite list of strict monomial comparisons can do so, by theorem 6.1. An order-based proof must use a uniform statement over unbounded exponents, a compactness principle retaining arithmetic data, or a separate exact identity.

8.7 Can fixed-order confluence repair a determinant constant?

Not through the entrywise multiplier divisibility alone. The bound in theorem 7.4 shows that such a gain is lower-order. A successful confluent determinant would need global cancellation, a growing jet regime, or an exact factorization invisible in termwise valuations.

9 Conditional and restricted positive results

The full conjecture remains beyond current unconditional methods, but several nearby statements are complete.

Theorem 9.1 (Three algebraic bases from Six Exponentials). *Let $a_1, a_2, a_3 > 1$ be algebraic numbers whose logarithms are linearly independent over $\mathbb{Q}$. If $x \in \mathbb{R}$ and all three a_i^x are algebraic, then $x \in \mathbb{Q}$.*

Proof. If $x \notin \mathbb{Q}$, then $1, x$ are $\mathbb{Q}$ -linearly independent. The Six Exponentials Theorem applied to the three logarithms $\log a_i$ and the two numbers $1, x$ says that at least one of the six exponentials a_i or a_i^x is transcendental, contrary to the hypotheses. $\square$

Corollary 9.2. *If $2^x, 3^x, 5^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Proof. The logarithms of 2, 3, 5 are $\mathbb{Q}$ -linearly independent by unique factorization. Apply theorem 9.1, then theorem 2.2. $\square$

Proposition 9.3 (A logarithm-product independence criterion). *Suppose the products $\log p \log q$, with p, q ranging over primes, are linearly independent over $\mathbb{Q}$ subject only to the symmetry $\log p \log q = \log q \log p$. Then theorem 1.1 holds.*

Proof. Factor hypothetical integer outputs as $M = \prod_p p^{a_p}$ and $N = \prod_q q^{b_q}$. Equation (3) becomes

$$\sum_p a_p \log p \log 3 - \sum_q b_q \log q \log 2 = 0.$$

Under the stated independence, the coefficient of every unordered prime pair must vanish. The only possible supports are then $M = 2^m$ and $N = 3^m$ with the same nonnegative integer m , giving $x = m$. $\square$

Remark 9.4. The hypothesis in theorem 9.3 is far stronger than any currently known theorem and is stated only to identify the exact type of algebraic independence that would settle the problem.

10 Precise research targets exposed by the new work

Research target 10.1 (Cross-family prime transfer). Find a theorem that uses the exact equality $\log N / \log M = \log 3 / \log 2$ to force a nontrivial relation between the prime supports of

$$3^{q_j} - 2^{p_j} \quad \text{and} \quad N^{q_j} - M^{p_j}$$

for infinitely many convergents. Real asymptotics and within-family gcd bounds are insufficient; the needed input must be genuinely local-global.

Research target 10.2 (Denominator confinement). Prove, from the counterexample hypotheses, that the reduced denominators of Q_j or the prime-to- M parts of the denominators of T_j lie in a fixed finite prime set or satisfy a recurrent divisibility law. Combined with the unimodular separation in theorems 5.2 and 5.6, such a result would create strong pressure toward eventual integrality.

Research target 10.3 (Leading-order jet cancellation). Construct a confluent determinant whose p -adic valuation exceeds the minimum valuation of its nonzero Leibniz terms by $cR^2 + o(R^2)$ for some $c > 0$, while preserving nonvanishing and a matching archimedean estimate. Theorem 7.1 shows that bounded entrywise integer multipliers alone cannot do this.

Research target 10.4 (Infinite order rigidity). Replace finite monomial comparisons by a uniform order property over an unbounded region and prove that the combination of that property with integrality characterizes the slope $\log 3 / \log 2$. Theorem 6.1 shows why any finite truncation is inadequate.

11 Literature check through August 21, 2026

The current literature check supports the same status conclusion as the source audit.

- (i) Waldschmidt’s chapter on the Four Exponentials Problem states the exact question whether 2^t and 3^t integral force t to be a nonnegative integer, and explains that a positive answer follows from Four Exponentials; the relevant complex and p -adic questions remain open.
- (ii) The 2026 Lean formalization of Gelfond–Schneider rigorously covers the algebraic-irrational case used in theorem 2.4. It does not cover transcendental exponents, which are exactly the remaining case.
- (iii) Recent Padé work on linear independence of polylogarithmic periods and products of polylogarithms develops powerful approximation machinery, but does not establish the logarithm-product independence required by (3).
- (iv) Recent work on additive relations in irrational powers concerns a different rigidity problem. It provides useful surrounding techniques but does not imply simultaneous integrality of the two powers only at integer exponents.

This report makes no global priority claim for its elementary lemmas. The novelty statement is limited and verifiable: the particular unimodular gap gcd package, rational secant bracket package, finite order-signature theorem, and fixed-order entrywise jet budget were not located in the attached source report in the forms proved here.

12 Conclusion

The requested implication has not been proved unconditionally here, and it would be mathematically false to present the partial results as a proof. The source audit and the current literature both place the problem at the Four Exponentials boundary. Nevertheless, the investigation produces four exact advances relative to the attached report:

- (1) determinant distance controls common divisors of exponential binomial gaps, giving adjacent coprimality along the continued fraction of $\log 3 / \log 2$;
- (2) a hypothetical counterexample transports these gaps into rational divided differences and alternating rational secant brackets for x , with sharply separated denominator support;
- (3) finite monomial order data cannot isolate the special logarithmic slope;
- (4) fixed-order integer jets are provably lower-order at the termwise budget level unless a new global cancellation mechanism is found.

The first two results create a concrete arithmetic interface between continued fractions and the hypothetical integer outputs. The last two eliminate broad classes of finite or local closing arguments. The remaining challenge is correspondingly narrow: one needs a mechanism coupling the real logarithmic identity to prime support or determinant cancellation at a scale that current linear-forms, Padé, automatic, and termwise p -adic methods do not reach.

A Detailed asymptotic bounds

For completeness, here is a nonasymptotic version of the estimates used in section 5. If $|u| \leq 1/2$, the mean value theorem gives

$$e^{-1/2} |u| \leq |e^u - 1| \leq e^{1/2} |u|.$$

Consequently, whenever $|\delta_j| \leq 1/(2x)$,

$$xe^{-1} 2^{p_j(x-1)} \leq Q_j \leq xe^{1/2} 2^{p_j(x-1)},$$

a deliberately coarse but uniform two-sided exponential estimate. More precise constants follow by applying Taylor's theorem with remainder to Φ_x on a compact interval around zero.

The continued-fraction error bound

$$\left| \alpha - \frac{p_j}{q_j} \right| < \frac{1}{q_j q_{j+1}}$$

immediately gives (17). Since the convergents alternate around an irrational number, δ_j alternates in sign, and the strict monotonicity of Φ_x converts this into the rational brackets of theorem 5.4 without any asymptotic qualification.

B Reproducible continued-fraction computation

The numerical entries in table 1 can be reproduced with arbitrary precision using the following short script.

```

import mpmath as mp
from math import gcd

mp.mp.dps = 200
alpha = mp.log(3) / mp.log(2)

# Continued-fraction coefficients.
a = []
y = alpha
for _ in range(9):
    z = int(mp.floor(y))
    a.append(z)
    y = 1 / (y - z)

# Convergents p_j / q_j.
p0, p1 = 0, 1
q0, q1 = 1, 0
conv = []
for z in a:
    p = z * p1 + p0
    q = z * q1 + q0
    conv.append((p, q))
    p0, p1 = p1, p
    q0, q1 = q1, q

gaps = [pow(3, q) - pow(2, p) for p, q in conv]
for j, ((p, q), g) in enumerate(zip(conv, gaps)):
    next_gcd = gcd(abs(g), abs(gaps[j+1])) if j+1 < len(gaps) else None
    print(j, p, q, g, next_gcd)

```

The computation is illustrative only. The coprimality statement itself is proved symbolically in theorem 4.6 and does not depend on numerical verification.

C Audit checklist for future claimed proofs

A proposed proof should be rejected or repaired if it contains any of the following transitions without a new theorem.

- (1) From $\det(\log a_{ij}) = 0$ to a rational dependence of rows or columns. This is the Four Exponentials step.
- (2) From real-logarithmic equality to congruence equality at a prime. Real and p -adic logarithms are not interchangeable.
- (3) From many integer values to automaticity, recurrence, or an algebraic generating function without constructing the required finite-state or functional structure.
- (4) From a termwise p -adic divisor to an extra leading determinant constant without charging the same multiplier in the archimedean bound.
- (5) From rational approximants to rationality of x without comparing the error to their reduced denominator height.
- (6) From exhaustive computation below a bound to a proof for all x .
- (7) From a conditional theorem under Four Exponentials, Schanuel, or an unproved logarithm-product independence statement to an unconditional conclusion.

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